

Anisotropic Multiexciton Quintet and Triplet Dynamics in Singlet Fission via Pulsed Electron Spin Resonance

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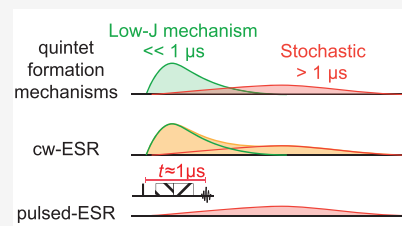


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ABSTRACT: The quintet triplet-pair state may be generated upon singlet fission and is a critical intermediate that dictates the fate of excitons, which can be exploited for photovoltaics, information technologies, and biomedical imaging. In this report, we demonstrate that continuous-wave and pulsed electron spin resonance techniques such as phase-inverted echo-amplitude detected nutation (PEANUT), which have emerged as the primary tool for identifying the spin pathways in singlet fission, probe fundamentally different triplet-pair species. We directly observe that the generation rate of high-spin triplet pairs is dependent on the molecular orientation with respect to the static magnetic field. Moreover, we demonstrate that this observation can prevent incorrect analysis of continuous-wave electron spin resonance (cw-EPR) measurements and provide insight into the design of materials to target specific pathways that optimize exciton properties for specific applications.



INTRODUCTION

Singlet fission (SF) is a fast, spin-conserving electronic process in which one high-energy singlet exciton decays into two low-energy triplet excitons.^{1,2} SF has attracted interest as a way to improve the efficiency of photodetectors and photovoltaics^{3,4} through energy downconversion and has also been investigated as a route to well-defined photogenerated spin states for use in quantum information^{5,6} or dynamic nuclear polarization.⁷ SF can occur in a wide range of materials including solutions,^{8–10} organic crystals,^{11–13} and semiconducting polymers.^{14,15} Molecular chromophore dimers^{9,16–19} have emerged as key platforms for mechanistic studies of SF due to their well-described structures and tunable photophysical parameters.^{16,19–21}

The triplet-pair multiexciton is a key intermediate in SF and contains three spin-manifolds, with multiplicity $M = 1, 3$, and 5 , respectively;^{22,23} we denote these manifolds by $^M(\text{TT})$. Singlet fission proceeds from a singlet exciton S_1 into the singlet multiexciton $^1(\text{TT})$ and can subsequently dephase into the $^3(\text{TT})$ (triplet) or $^5(\text{TT})$ (quintet) manifolds;^{9,24–26} dissociate to free triplets; or annihilate to a single singlet or triplet exciton. As direct decay of $^5(\text{TT})$ to the singlet ground state is doubly spin-forbidden, quintet formation could reduce losses in SF-sensitized photovoltaics.²⁷ SF-derived quintets have also attracted attention as five-level organic qubits that can be photoinitiated into orientation-defined spin states by optical pumping, satisfying the di Vincenzo criteria for quantum information processing.⁵ These desirable properties make it vital to understand the behavior and dynamics of the $^5(\text{TT})$ state and its role in SF.

The $^5(\text{TT})$ quintet and $^1(\text{TT})$ singlet are separated in energy by 3 times the exchange energy J , corresponding to the electronic overlap between the triplet excitons. Here we assume that this intertriplet exchange interaction is dominated by its isotropic component; see previous works^{9,28,29} for the corresponding spin Hamiltonian. Quintets are only well-defined states when J is large in magnitude relative to the zero-field splitting energies (D and E). However, if J is large and time-independent, quintets cannot form from singlets. Instead, quintet formation is believed to occur via an exchange that fluctuates over time.³⁰ Our previous work suggests two qualitatively different pathways to SF quintets, enabled by different $J(t)$ dynamics, which yield different sublevel populations of quintets (Figure 1a). In what we here describe as “low- J ” quintet formation, $J(t)$ approaches 0 and allows coherent singlet–quintet mixing until an increase in J projects the mixed state onto $^5(\text{TT})$ (Figure 1b).²⁸ While $J \approx 0$, coherent mixing occurs at a rate of $D \approx 10^9 \text{ s}^{-1}$, and this pathway can thus lead to quintet formation within nanoseconds. If $J(t) \geq 0$ (as is generally the case in low- Z molecular materials), at high magnetic fields the $m_s > 0$ $^5(\text{TT})$ sublevels remain separated from $^1(\text{TT})$ by the Zeeman interaction, and this pathway produces quintet populations biased toward $m_s \leq 0$. If $J(t)$ continues to fluctuate between large and near-zero values

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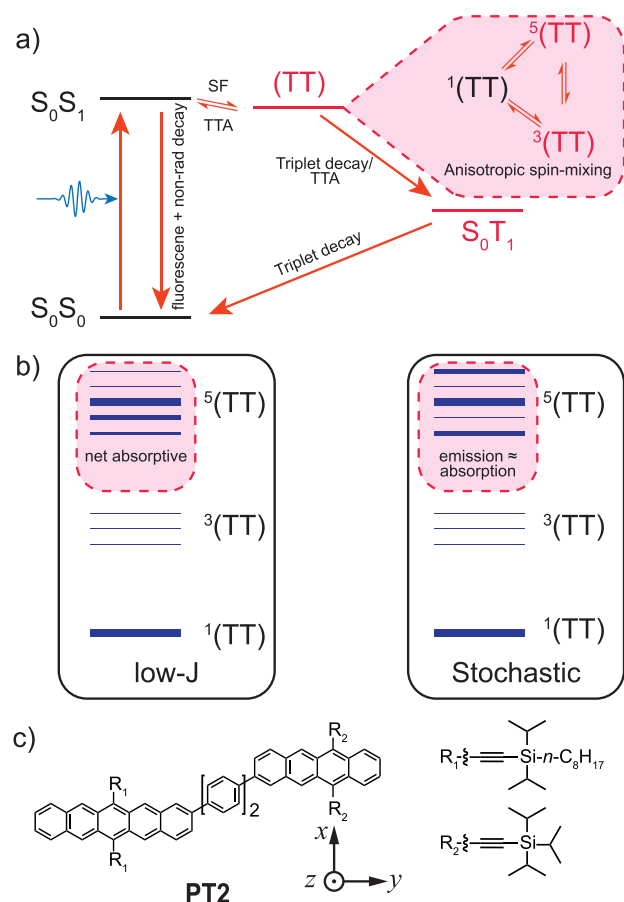


Figure 1. Singlet fission in PT2. (a) The photophysical processes in PT2. The states that are observable in spin resonance measurements are marked in red. (b) Schematic of the predicted TT occupancy for triplet pairs governed by stochastic or low- J spin-mixing. (c) The structure of PT2, the intramolecular singlet fission material used in this study. R_1 is triisopropylsilyl ethynyl (TIPS), and R_2 is n -octyldiisopropylsilyl ethynyl (NODIPS).

following photoexcitation, repeated mixing cycles may provide a relaxation pathway for $^5(\text{TT})$ spins and limit coherence times.

An alternative to low- J quintet formation is what we here describe as “stochastic” quintet formation, where $J(t)$ fluctuates while remaining large relative to D .³⁰ As J is large at all times, coherent $^1(\text{TT}) \leftrightarrow ^5(\text{TT})$ mixing does not occur. Instead, each sharp change in J has a low probability of driving diabatic Landau–Zener-type transitions from $^1(\text{TT})$ to $^5(\text{TT})$.³¹ As this is a slower process than the coherent mixing of low- J quintets, stochastic quintets are expected to form more slowly but possess longer spin coherence times in the absence of relaxation through singlet–quintet mixing. Mixing to the $^3(\text{TT})$ manifold may also occur in systems where chromophore zero-field splitting axes are not parallel.³² While fluctuating exchange plays a key role in theoretical models of quintet formation, direct experimental measurements of exchange are challenging and thus far limited to single-crystal materials.^{33,34}

Regardless of the mechanism, quintet formation in a magnetic field is inherently anisotropic across molecular orientation. This arises because only anisotropic elements of the spin Hamiltonian do not commute with the S^2 operator. For many SF materials, this anisotropy is mainly due to the intratriplet magnetic dipole, which is represented by the zero-field splitting term. In general,

anisotropies in the Zeeman, intertriplet dipole–dipole coupling, and hyperfine terms also contribute to this effect.³⁵

The quintet and triplet multiexcitons formed in SF possess similar overall energies but different spin characteristics. Electron spin resonance (ESR) has emerged as a powerful spectroscopic tool for studying these transient electronic states.^{36,37} Standard continuous-wave ESR (cwESR) experiments measure the field-dependent absorption or emission of microwaves following pulsed laser photoexcitation, but where multiple spin states coexist this picture can be complicated by the multiple overlapping spin transitions. Postprocessing approaches such as spectral deconvolution and kinetic modeling are required to resolve the summed cw-ESR spectrum into its corresponding transitions, but these techniques require many fitting parameters.

The difficulty of cwESR analysis is partly due to the large amount of information that is contained in a cwESR spectrum. Under high-field conditions, the zero-field interactions act as perturbations upon the high-field states. As a result, the magnetic field strength at which resonance occurs can be mapped to the direction of the magnetic field, $\mathbf{B}$, with respect to the molecular axes (colored lines and dots in Figure 2a).

The overall effect is that each spin transition of a single molecular orientation is resonant at a different magnetic field, while experimental cwESR measurements at a single magnetic field represents the sum of multiple unknown spin transitions at multiple molecular orientations. As a result, a given population of randomly oriented evolving spins will generate a unique time-dependent cwESR spectrum, but a given time-resolved ESR spectrum can be explained by multiple combinations of orientationally anisotropic spin parameters. This makes it challenging to interpret cwESR spectra involving randomly oriented spin systems with access to multiple spin transitions.

We combine two time-resolved ESR techniques to deconvolve this anisotropy and report on the relative rates of quintet formation (Figure 2b,c). This allows us to (i) identify anisotropic triplet generation for the first time in SF and (ii) distinguish quintets formed via the “low- J ” mechanism/pathway from those formed via the “stochastic” mechanism. Pentacene and/or tetracene dimers are well-known intramolecular SF materials,^{16–18,20,21} and we turned to the previously reported compound PT2 (Figure 1c) as a model substrate for nutation-resolved pulsed ESR studies. This heterodimer is chosen so that the triplet states can be unambiguously identified to reside on either the tetracene or pentacene subunit as a result of their differing zero-field splitting parameters. SF in this heterodimer has been shown to yield long-lived triplet states that persist for approximately 100 μs ,⁴⁰ which we judged suitable for resolution by 2D nutation spectroscopy at ESR time scales.

RESULTS AND DISCUSSION

Noncoherent Spectroscopy: cw-ESR. Transient cw-ESR of PT2 in glassy toluene at 50 K (Figure 3) shows four major peaks at 328, 342, 354, and 367 mT. As cw-ESR cannot attribute spectral features to specific transitions, these transitions can only be tentatively assigned by analogy to prior work.^{9,24}

The distinctive $\sim(D - 3E)/3$ zero-field splitting makes an assignment of the “inner peaks” (342 and 354) to the $^5(\text{TT})_{0 \leftrightarrow \pm 1}$ transition relatively unambiguous. We are able to estimate the intertriplet dipole–dipole coupling from the deviation from $(D - 3E)/3$ and find this to be 50 MHz (assuming tetracene triplet $D = 1300$ MHz²⁴ and pentacene triplet $D = 1138$ and $E = 19$ MHz⁹), in close agreement with n -octyldiisopropylsilyl ethynyl

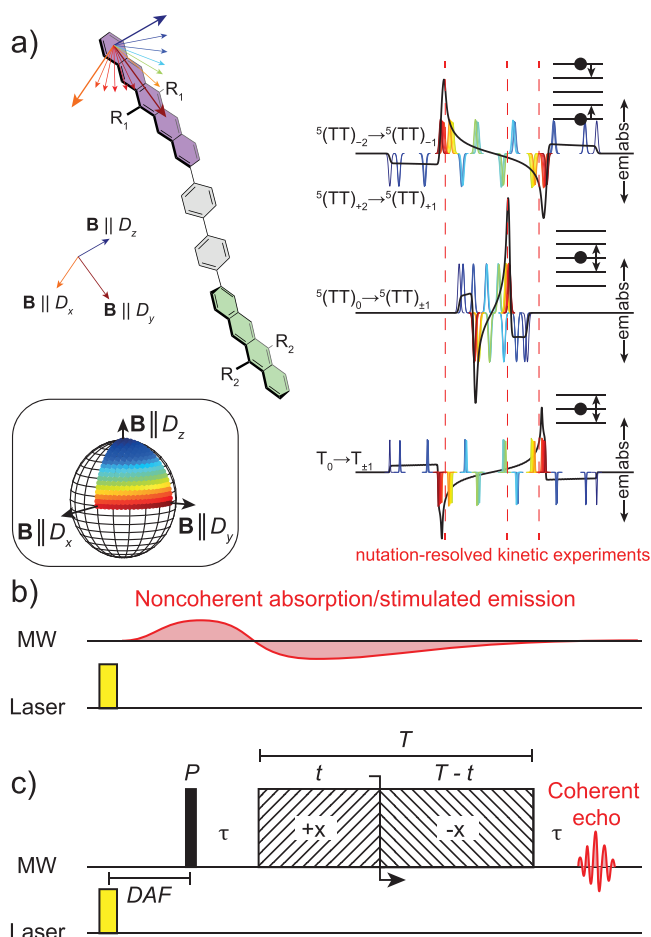


Figure 2. Unravelling spin multiplicity and anisotropy using spin resonance spectroscopies. (a) Simulated powder spectra of spin-polarized species (black lines) are the sum of the individual transitions shown in colored lines. The colors of these lines are mapped to the direction of the external magnetic field with respect to the molecular axes (insets). (b) Transient continuous-wave ESR (cw-EPR) uses a weak continuous-wave microwave signal to measure the absorption and/or stimulated emission of transient spin species. (c) PEANUT³⁸ pulsed ESR (p-EPR) sequence uses a long composite pulse to generate a coherent rotary echo and measures spin nutation as a function of the phase inversion point.

(NODIPS)–pentacene dimers, which also have a biphenyl covalent bridge.⁹ Note that, in order to extract these D and E values, we have assumed that the triplet excitons are localized on each chromophore. This suggests that the biphenyl bridge is not coplanar with the tetracene or pentacene chromophores. However, as shown in Section 3.3 of the SI, the dihedral angle between the pentacene and tetracene chromophores is small. As such, we assume the biphenyl bridge is not coplanar with these chromophores, as depicted in Figure 2a and in accordance with ref 41. The absorptive “lump” feature at the center of the spectrum (e.g., Figure 3b at 0.1 μ s) is indicative of quintets with population-averaged spin quantum number $\overline{m}_s < 0$, suggesting quintet generation via the low- J mechanism.

However, the nature of the “outer peak” signals (328 and 367 mT) split by $(D - 3E)$ remains less clear. Both $^3(T)_{0 \leftrightarrow \pm 1}$ and $^5(TT)_{\pm 1 \leftrightarrow \pm 2}$ transitions would occur at these resonant fields. The $^3(TT)_{\pm 1 \leftrightarrow \pm 1}$ transitions are also resonant at these fields, but we make the assumption that the z -axes of the pentacene and tetracene chromophores are approximately parallel (to within

10° between the planes of the chromophores (see Supporting Information Section 3.3)), and, therefore, we do not expect coupling to the $^3(TT)$ to be competitive with the $^5(TT)$ manifold.²⁸

On close examination of spectra, we note a few indications of anisotropy of the evolving spin dynamics (anisotropy here regarding the orientation of the molecular zero-field splitting tensor axes to the magnetic field). First, we note a weak signal at the edges of the spectra within 0.1 μ s (see circled, shaded areas in Figure 3b). This can only be due to the outer $^5(TT)$ or the uncoupled triplets with the magnetic field aligned along the z -axis of the molecule. However, we do not observe the expected intensity at the x, y peaks (at 328 and 367 mT). This suggests that triplet excitons form more slowly in molecules with a zero-field splitting (ZFS) tensor aligned perpendicular to the external field ($D_{z \perp B}$) and more quickly in parallel molecules ($D_{z \parallel B}$). This minor feature has hitherto been ignored, but more clarity is provided by the phase-inverted echo-amplitude detected nutation (PEANUT) data (*vide infra*). We find evidence for at least four species using singular value decomposition and are able to fit the data using a kinetic model; however, as we will show using the PEANUT data below, there are anisotropic effects playing a major role in these materials. While we are able to capture triplet decay anisotropy in a kinetic model, due to the overlap between the triplet and $^5TT_{\pm 1 \leftrightarrow \pm 2}$ transitions, the true dynamics of the system can be obfuscated by this approach. An example model that demonstrates this is provided in Section 9 of the SI. As cw-EPR spectroscopy (Figure 3) cannot tell us whether $^5(TT)_{\pm 2}$ quintet excitons are present in this system, we turn to pulsed ESR (p-EPR) nutation spectroscopy to resolve and identify the underlying spin transitions.

Coherent Spectroscopy: p-EPR Nutation. Rabi nutation pulsed ESR (p-EPR) experiments allow the identification of unknown spin transitions by the characteristic relationship between nutation frequency ω_{nut} and the spin quantum numbers of the transition:

$$\omega_{\text{nut}} \propto \sqrt{S(S+1) - m(m+1)} \quad (1)$$

where S is the total spin of the manifold (e.g., $S = 2$ for quintet transitions) and m_s is the lower of the two spin projections involved in the transition.

Nutation ESR studies of SF spin states^{9,24} typically use the Hahn-echo pulse sequence. In this experiment, nutation measurements are collected as a damped oscillating series of measurements while incrementing the length of polarizing pulse P . The resulting signals are thus acquired with nonconstant total delivered energy and potentially experience different relaxation dynamics.⁵ Here, we instead use the constant-time, constant-power PEANUT pulse sequence³⁸ (Figure 2c) to eliminate these concerns and separate coherent nutation from the other dynamics of the evolving SF-derived states.^{42,43}

Previous nutation studies of SF have generally acquired single nutation measurements at specific points in magnetic field and time and used these to qualitatively identify the presence or absence of different spin transitions. Here, we extend these one-dimensional measurements of nutation frequency into a second dimension by collecting multiple full nutation spectra using the PEANUT sequence³⁸ at a series of magnetic fields or times. Following Fourier transformation along the nutation axis, we obtain 2D spectra correlating nutation frequencies to magnetic field or time.

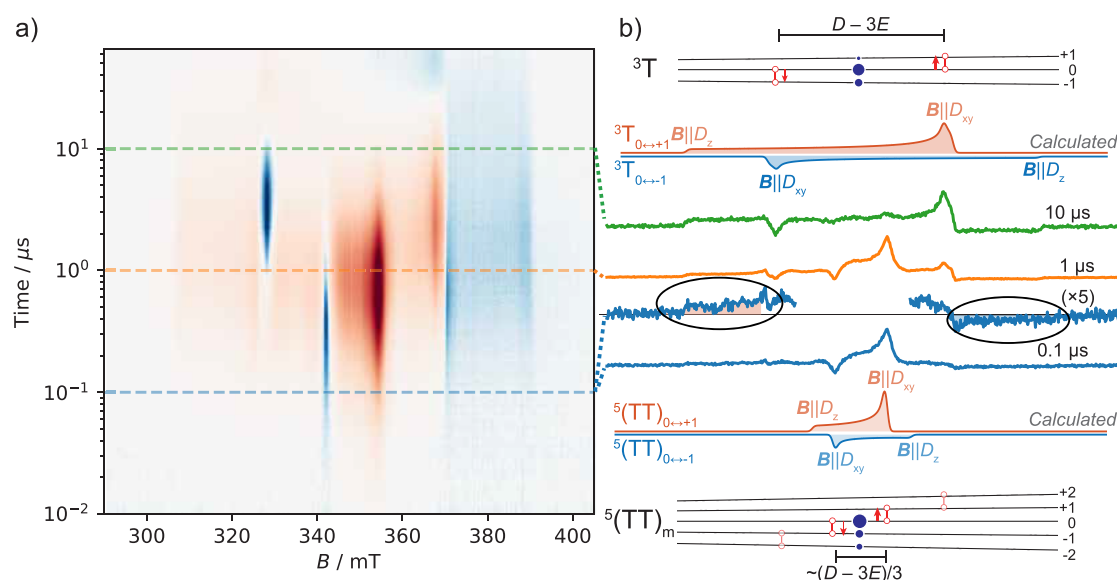


Figure 3. Noncoherent transient cw-EPR spectrum of PT2 at 9.71 GHz, toluene, 50 K. (a) ${}^5(\text{TT})_{0\leftrightarrow\pm 1}$ “inner” quintet transitions are prominent at 343 and 354 mT shortly after photoexcitation but are subsequently replaced by triplet signals at 325 and 365 mT. (b) Extracted spectra at 0.1, 1, and 10 μs . These spectra are dominated by the ZFS interaction, characterized by axiality D and rhombicity E , with resonances at different fields corresponding to different orientations of the spin system to B . When the ZFS tensor is approximately axisymmetric ($E \approx 0$), a “Pake pattern” forms with peaks corresponding to the many spins orientated perpendicular to the external field (Figure 2a). These $D_z \perp B$ transitions are split by $D - 3E$ for the ${}^3(\text{T})_{0\leftrightarrow\pm 1}$ triplet and ${}^5(\text{TT})_{\pm 1\leftrightarrow\pm 2}$ “outer” quintet transitions, while the “inner” quintet transitions are more closely separated by $(D - 3E)/3$ (calculated powder spectra in red and blue using EasySpin³⁹). The pattern of absorptive and emissive peaks shows that these spins are populated with an excess of $m_s = 0$ spins.

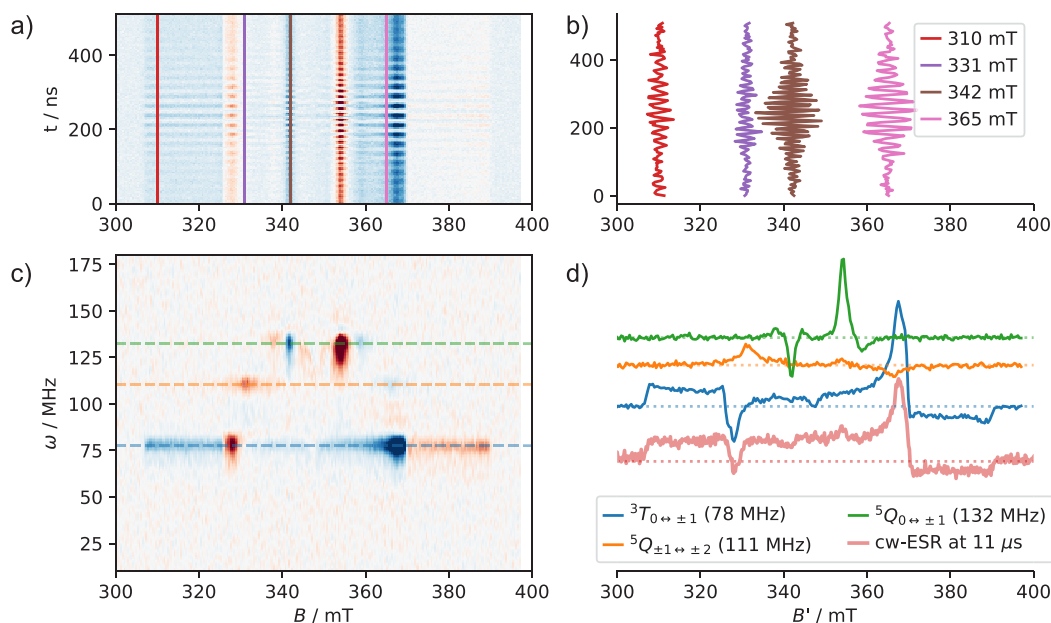


Figure 4. Coherent field-swept 2D PEANUT nutation of PT2 at 11 μs . (a) Raw data showing oscillatory signals, with (b) selected time-domain nutation traces shown. (c) Frequency-domain data obtained as the Fourier transform of (a). Signals at 78, 111, and 132 MHz fall at a ratio of 1, $\sqrt{2}$, and $\sqrt{3}$ and can be assigned to ${}^3(\text{T})_{0\leftrightarrow\pm 1}$, ${}^5(\text{TT})_{\pm 1\leftrightarrow\pm 2}$, and ${}^5(\text{TT})_{0\leftrightarrow\pm 1}$ transitions (see eq 1). (d) Transition-selective spectra extracted from (c) as horizontal slices, compared to cw-EPR data from Figure 3. The different nutation frequencies of the transitions mean that the single polarization pulse cannot polarize all spins equally, and the amplitude and sign of these signals should not be compared directly (see Section 4 of the SI.).

Field-Swept Transition Spectra Resolved by p-EPR Nutation. To conclusively identify the transitions contributing to the previously obtained cw-EPR data (Figure 3), we first performed a field-swept 2D nutation experiment on PT2 at 11 μs , shown in Figure 4 (9.73 GHz, 50 K, toluene, 590 nm irradiation). The data as-acquired are given in 4a, with nutation clearly visible as an oscillating signal along the y-axis. The same

data after Fourier transformation in the nutation axis are shown in Figure 4c, with signals visible at nutation frequencies of 78, 111, and 132 MHz. The ratio of $1.69 \approx \sqrt{3}$ between the highest and lowest nutation frequencies allows these to be assigned to the ${}^3(\text{T})_{0\leftrightarrow\pm 1}$ ($\omega = 78$ MHz) and ${}^5(\text{TT})_{0\leftrightarrow\pm 1}$ ($\omega = 132$ MHz) transitions, consistent with eq 1 and with previous studies of SF. The middle signal at $\omega = 111$ MHz occurs at a frequency of 1.42

$\approx \sqrt{2}$ relative to the triplet nutation, and can therefore be assigned to the $^5(\text{TT})_{\pm 1} \leftrightarrow \pm 2$ “outer” quintet transitions. Slicing the 2D field-nutation spectrum in Figure 4b at these frequencies allows us to obtain field-dependent spectra resolved by each of these transitions, which are shown in Figure 4d alongside the cw-ESR spectrum at a corresponding time.

We interpret the nutation data as suggesting that the $^5(\text{TT})$ quintet is initially polarized with an excess of $m_s = 0, \pm 2$ over the $m_s = \pm 1$ spin states (for the major peaks at $B \perp D_z$). The presence of a weak signal at 367 mT assigned to the $^5(\text{TT})_{+2}$ state is not compatible with the low- J mechanism, but is suggestive of the stochastic mechanism³⁰ (see the sublevel populations in Figure 1b). The deviation of the spectral line shape from a classic Pake pattern also suggests that $m_s = \pm 2$ quintets form predominantly at orientations close to $B \perp D_z$ and not at $B \parallel D_z$, consistent with the matrix element overlap given by $\langle ^1(\text{TT})_0 | \hat{H} | ^5(\text{TT})_{\pm 2} \rangle$ (see Section 8.2 of the SI). We therefore attribute the circled signal in Figure 3b to triplet formation for molecules with the magnetic field aligned along the z -axis of the molecules. Additional field-swept nutation spectra taken at 3 and 16 μs (see Section 5 of the SI) show the presence of strong $m_s = \pm 2$ quintet signals at 3 μs after flash and a weak residual signal at longer time periods in a spectrum otherwise dominated by long-lived free triplets.

Time-Swept Kinetics Resolved by p-ESR Nutation. This observation of transient $^5(\text{TT})_{\pm 2}$ quintets in PT2 demonstrates that the spin dynamics of SF are more complex than a linear evolution from $^1(\text{TT})_0$ singlets to $^5(\text{TT})_0$ quintets to free triplets⁶ and raises the question of how these transient exciton populations evolve over time. Previous studies⁹ have investigated the kinetics of SF excitons by fitting kinetic-spectral models to transient cw-ESR data, but these multiparameter fits are rarely unambiguous and rely on correct kinetic models of the underlying dynamic populations. To avoid the challenging deconvolution of kinetics and unknown spectra from transient ESR data, we again turn to coherent nutation ESR and the PEANUT experiment.

The long measurement time required to acquire a complete 2D field-swept nutation spectrum at a single time point (Figure 4) makes it impractical to collect full 3D spectra as a function of nutation, field, and delay time. Instead, we acquire series of time-resolved nutation spectra at a fixed magnetic field while systematically varying the delay after laser flash. As with the field-swept spectra in Figure 4, Fourier transformation of the 2D PEANUT data along the nutation axis yields spectra resolved by transition, but we now resolve the intensity of each transition as a function of time rather than field.

Figure 5 shows time-swept PEANUT nutation-resolved kinetic traces at 331 and 366 mT, the fields corresponding to the maximum intensities of the $^5(\text{TT})_{\pm 2 \leftrightarrow \pm 1}$ peaks in Figure 4. In addition to the $^5(\text{TT})_{\pm 2 \leftrightarrow \pm 1}$ transition, these fields are close to resonance for the $T_0 \leftrightarrow \pm 1$ triplet transitions oriented with $B \perp D_z$ but are outside the range of expected $^5(\text{TT})_{0 \leftrightarrow \pm 1}$ transitions (see Figure 3b). As such, we also performed a PEANUT experiment at 342 mT to capture the $^5(\text{TT})_{0 \leftrightarrow \pm 1}$ transition dynamics for $B \perp D_z$.

We apply the model in Figure 5c) to fit the data. Since $B \perp D_z$ for all 5 traces in Figure 5a,b), we fit the data simultaneously. The kinetic equations are provided in Section 10 of the SI. We model population transfer via the stochastic mechanism from $^1(\text{TT})$ to $^5(\text{TT})_0$ at all orientations. Further $^1(\text{TT})$ to $^5(\text{TT})_{\pm 2}$ population transfer occurs when the applied magnetic field is perpendicular to D_z . Each quintet may decay to a single

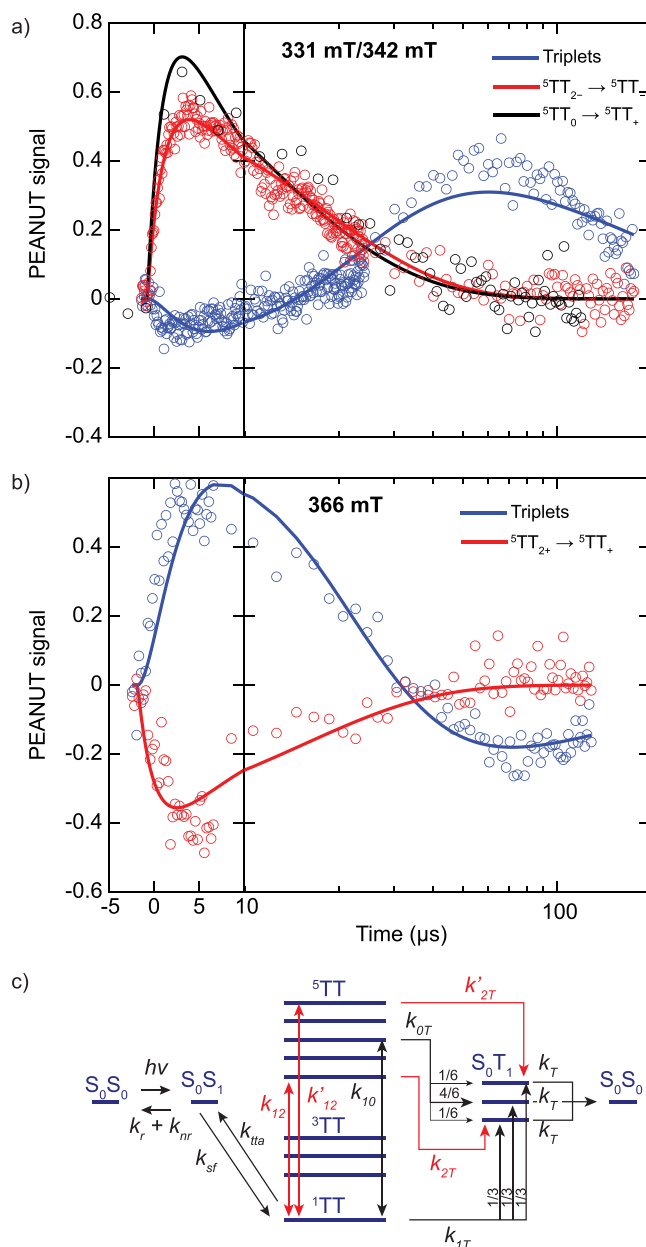


Figure 5. Nutation-resolved kinetics of PT2 excited states. Time-swept PEANUT nutation spectra showing transitions obtained at (a) 331 mT (342 mT for the $^5\text{TT}_{0 \leftrightarrow \pm 1}$ transition) and (b) 366 mT. (c) The rate model used to fit the data in (a) and (b). Red arrows denote anisotropic transitions that are faster at $B \perp D_z$ orientations.

pentacene triplet. The ratio of triplet sublevel populations is governed by the quintet eigenstates, which we express in the high-field triplet basis $|-\rangle$, $|0\rangle$, and $|+\rangle$. That is, $|^5\text{TT}_0\rangle = \frac{1}{\sqrt{6}}(2|00\rangle - |+-\rangle - |-+\rangle)$ produces triplets $|-\rangle$, $|0\rangle$, $|+\rangle$, with a ratio 1/6, 4/6, 1/6. The $|^5(\text{TT})_{-2}\rangle$ and $|^5(\text{TT})_{+2}\rangle$ exclusively populate the $|-\rangle$ and $|+\rangle$ pentacene triplet sublevel, respectively.

The quintet to single pentacene triplet rates have two potential explanations, both of which can be modeled using the scheme shown in Figure 5c. The first is simply that one triplet decays, leaving the other behind. If the remaining triplet exciton exists on the tetracene chromophore, it will undergo rapid Dexter transfer to the pentacene chromophore, in accordance with our detection of pentacene triplets exclusively.

The second possibility is that this process occurs via the $^3(\text{TT})$ manifold, as previously predicted.^{26,27,44} We do not observe any evidence for the $^3(\text{TT})$ manifold in the field-swept PEANUT data (see SI Section S5). While the nutation frequency of $^3(\text{TT})$ transitions would be identical to S_0T_1 , we would expect a slightly different spectrum due to intertriplet dipolar coupling. However, if the $^3(\text{TT})$ state is short-lived, this would not be observed in the ESR measurements and the spin polarizations of the quintet would be retained. This is also reasonable given that the (TT) energy is close to the T_2 energy of pentacene, making triplet–triplet annihilation from $^3(\text{TT})$ to S_0T_2 , followed by rapid internal conversion to S_0T_1 , a distinct possibility.²¹

The triplet signals for magnetic fields aligned along the x,y plane of the D tensor exhibit complex behavior, first appearing with one sign (emissive/absorptive) before passing through zero to give a signal of the other sign. Notably, this is not observed in the wings; that is, for magnetic fields aligned along the z -axis of the molecule there is no inversion in the signal. This population inversion was previously predicted to result from 3T_0 triplets decaying more quickly than $^3T_{\pm 1}$ triplets:^{9,35} as ESR can only measure the difference in population (“polarization”) between two states, this population inversion was suggested to yield an inversion of signal (Figure 5b,c inset).

However, the PEANUT data in this work provide new clarity to this feature. Since we have unambiguously demonstrated that the $^5(\text{TT})_{\pm 2}$ quintets are populated, but only for molecules with the field aligned along the x,y plane of the molecular D tensor, their decay into free triplets must be taken into account. As outlined in Figure 5c, the $^5(\text{TT})_{\pm 2}$ will only occupy the $T_{\pm 1}$ upon decay and will cause a sign inversion of the triplet signal. That is, it is not just anisotropic decay of triplets that causes the inversion in sign; anisotropic quintet formation leads to anisotropic generation of triplets. This is an important point, as we have previously observed signal inversion at later times in over five different singlet fission materials;^{9,17,19} this suggests that anisotropic $^5(\text{TT})_{\pm 2}$ generation is a somewhat ubiquitous feature.

The combined analysis of the PEANUT data is as follows. For molecules with the B -field parallel to D_z , singlet fission produces the $^5(\text{TT})_0$ state, which, upon decay to a single pentacene triplet, provides an overpopulation of T_0 compared to $T_{\pm}$. This is evidenced by the “wings” in the field-swept triplet spectrum (blue trace in Figure 4c). For this orientation, we do not observe any transitions due to population of the $^5\text{TT}_{\pm 2}$ state. That is, only the transitions marked with black arrows in Figure 5c are active.

On the other hand, for molecules oriented with the B -field perpendicular to D_z , transitions marked red in Figure 5c are active. This means that $^1(\text{TT}) \rightarrow ^5(\text{TT})_0$ and $^1(\text{TT}) \rightarrow ^5(\text{TT})_{\pm 2}$ are competing pathways for quintet generation, the latter of which provides an overpopulation of $T_{\pm}$ compared to T_0 .

In stark contrast to the cw-ESR spectra (net-absorptive, see Figure 4d) the PEANUT data appear to have approximately the same amount of absorptive and emissive signal. This suggests that quintets (and the resulting triplets) are produced via stochastic fluctuations in the intertriplet exchange energy (see Figure 1b).

Reconciling Coherent and Noncoherent Spectroscopy. There are clear differences between the cw-ESR signals and p-ESR nutation traces at the same fields. The noncoherent cw-ESR and coherent p-ESR experiments give two related but distinct views of the transient exciton dynamics of SF in PT2 (Figure 6a,b). This arises because of the difference in sensitivity

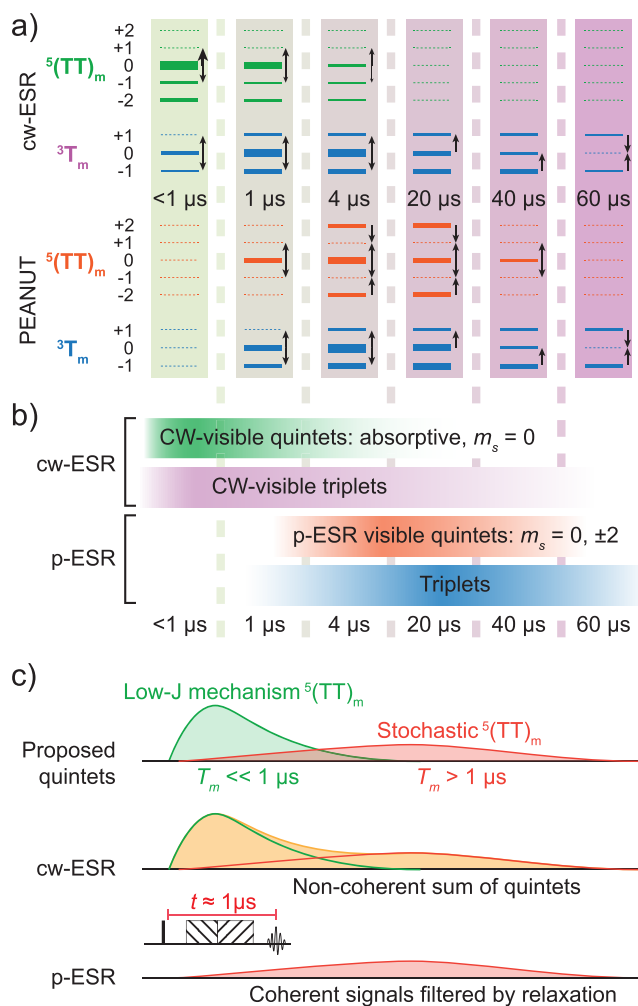


Figure 6. Coexisting routes to quintet formation in singlet fission. (a, b) Measurements by cw-ESR and nutation p-ESR find qualitatively different spin populations and dynamics. cw-ESR quintets are strongly net-absorptive and with faster kinetics, while p-ESR-resolved quintets have slower kinetics and are less net-absorptive. (c) These observations are consistent with coexisting nonstationary and stationary (Figure 1) routes to quintet formation, with the prompt nonstationary quintets filtered out of p-ESR by their shorter coherence times. Quintet populations indicated are for the $B \perp D_z$ orientation; see Sections 8.2–8.4 of the SI for details.

of the two techniques. p-ESR will only detect spin states that remain coherent for a time T , as shown in 2c. By contrast, cw-ESR has no coherence requirements.

As in previous studies of SF in weakly coupled molecular dimers, transient cw-ESR (Figure 3) suggests a direct progression from prompt $^5(\text{TT})_0$ quintets at $0.1\text{--}1 \mu\text{s}$ to longer lived triplets that persist for tens of μs . The overall spectrum is strongly absorptive, indicating an excess of $m_s < 0$ quintet excitons over $m_s > 0$ quintets. This rapid formation of predominantly absorptive quintets is consistent with the low- J quintet pathway²⁸ shown in Figure 1b.

In contrast, coherent p-ESR nutation data resolve a quintet population that forms and decays more slowly, is less net-absorptive, and includes comparable signals from both $^5(\text{TT})_{+2}$ and $^5(\text{TT})_{-2}$ quintets.¹⁰ The prompt component of the cw-ESR that rises and falls within $1 \mu\text{s}$ is entirely absent from the p-ESR data. Notably, the triplet signals resolved by p-ESR (Figure 4c) are strongly absorptive: as triplet formation is spin-conserving,

this suggests that p-ESR resolves the triplets formed from “prompt” quintets even though it does not resolve the quintets themselves. The overall combination of slow dynamics and the formation of roughly equivalent populations of $^5(\text{TT})_{+2}$ and $^5(\text{TT})_{-2}$ quintets is consistent with the stochastic quintet pathway³⁰ as shown in Figure 1b.

These distinct pictures of quintet dynamics obtained by cw-ESR and p-ESR can be reconciled by the different selectivities of these experiments to spin coherence times. Noncoherent cw-ESR measures “instantaneous” absorption/emission of microwaves and is relatively insensitive to coherence times, while p-ESR experiments detect only the spins that retain coherence through a pulse sequence (here approximately 1 μs). The prompt cw-ESR and delayed p-ESR signals (Figure 5) thus represent spin populations with short and long coherence times, respectively. While an overall increase in the spin coherence time of SF-generated quintets has been reported recently,⁵ the qualitatively different spectra obtained by p-ESR and cw-ESR suggest that this is *not* an overall increase in coherence time across a uniform spin population; rather, our data suggest the coexistence of different quintet subpopulations with different coherence times (Figure 6c) and different mechanisms of formation. These different quintet m_s populations are consistent with our theoretical models^{28,30} of quintet formation by fluctuating exchange where low- J quintets form quickly with short coherence times and net-absorptive polarization, while stochastically formed quintets are generated slowly with longer coherence times and different population sublevels (see Figure 1).^{28,30}

This analysis suggests that quintets formed via the stochastic mechanism have longer coherence times than those formed via the low- J mechanism. This is a reasonable result. The quintet manifold in the stochastic mechanism is always at a higher energy than the singlet state; that is, $J \gg D$ at all times. As a result, the spin multiplicity of these states is always well-defined.³⁰ By contrast, the low- J mechanism requires that $J \sim D$ at some point in time. This randomly drives singlet–quintet mixing, which leads to decoherence of the multiexciton state.

We arrive at two potential explanations for the coexistence of the two quintet formation mechanisms. The first is that there exist geometries of isolated molecules in a matrix where one mechanism is more likely than the other. For instance, a geometry that allows for rotation of the conjugated interchromophore bridge⁴⁵ would provide a pathway to a low- J configuration,²⁸ while molecules with a rigid bridge would retain a high exchange coupling, and quintet formation would proceed via the stochastic mechanism.

The second possibility is that there may be aggregation in our sample. Every effort was made to avoid aggregation, including synthesizing a new molecule with long NODIPS solubilizing groups and using low concentrations that show no evidence of aggregation at room temperature. However, if aggregation occurs, intermolecular triplet hopping is possible. Hopping provides a pathway to a low- J configuration.¹⁰ In this scenario we would expect cwESR data to observe aggregates and isolated dimers, but PEANUT would only observe isolated dimers.

As such, we suggest that the PEANUT experiment provides selectivity for isolated dimers that are not aggregated and have triplet pairs which remain coherent for longer times. This interpretation is consistent with a quintet population generated via the stochastic pathway, which always has large intertriplet exchange. This interpretation is also consistent with the recent work by Jacobberger et al. on single crystals, who demonstrated

longer triplet decay times in pulsed ESR measurements, as opposed to cwESR.⁵ Moreover, net absorptive cwESR signals have been observed in single crystals and thin films, which is indicative of hopping-mediated quintet formation, as expected.^{5,10,24}

CONCLUSION

We used coherent ESR nutation spectroscopy to resolve the transient spin states that arise during singlet fission, revealing different spin- and orientation-dependent populations by coherent and noncoherent ESR spectroscopy. These results point to multiple coexisting pathways to quintet formation in intramolecular SF materials and further demonstrate the power of 2D p-ESR experiments to probe complex and evolving spin systems.

In contrast to previous works invoking the anisotropic decay of triplets to describe flips in polarization, our data suggest that this is better explained by the anisotropic generation of quintet multiexcitons which feed triplet populations.

These findings support recent theoretical models of SF quintet formation by fluctuating exchange coupling, with dynamic regimes we here describe as “low J ”²⁸ ($J(t) \geq 0$) and “stochastic”³⁰ ($J(t) \gg D$). These qualitative regimes are consistent with populations of spins that are distinguishable via orientation and m_s and different coherence times: here, noncoherent cw-ESR measurements appear to reflect predominantly “low- J ” quintets, while coherent p-ESR measurements find a population more reminiscent of “stochastic” quintet formation. Understanding these competing formation pathways is critical, as judicious choice of the flavor of the quintet is likely to be necessary given the varied demands of different technological applications, e.g., quintets with long phase coherence times for information applications vs rapid triplet formation for energy applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.3c02672>.

Experimental methodology, data analysis details, additional ESR data, quintet formation simulations, synthesis details, and NMR spectra (PDF)

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Notes

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